

16 Chapter Review

➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

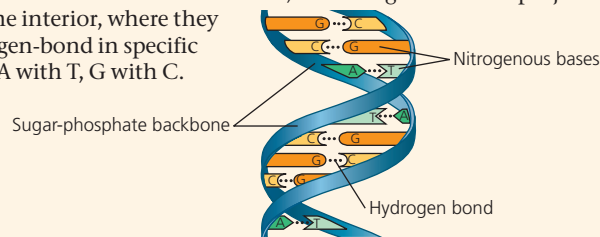
SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

CONCEPT 16.1

DNA is the genetic material (pp. 315–320)

- Experiments with bacteria and with **phages** provided the first strong evidence that the genetic material is DNA.
- Watson and Crick deduced that DNA is a **double helix** and built a structural model. Two **antiparallel** sugar-phosphate chains wind around the outside of the molecule; the nitrogenous bases project into the interior, where they hydrogen-bond in specific pairs, A with T, G with C.

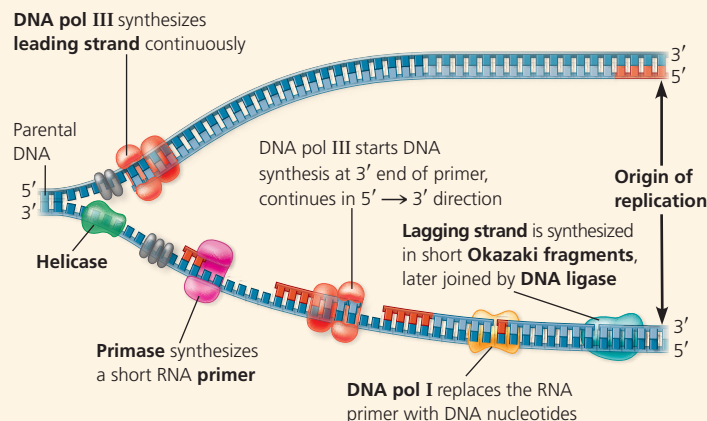


? What does it mean when we say that the two DNA strands in the double helix are antiparallel? What would an end of the double helix look like if the strands were parallel?

CONCEPT 16.2

Many proteins work together in DNA replication and repair (pp. 320–329)

- The Meselson-Stahl experiment showed that **DNA replication** is **semiconservative**: The parental molecule unwinds, and each strand then serves as a template for the synthesis of a new strand according to base-pairing rules.
- DNA replication at one **replication fork** is summarized here:



- DNA polymerases** proofread new DNA, replacing incorrect nucleotides. In **mismatch repair**, enzymes correct errors that persist. **Nucleotide excision repair** is a process by which **nucleases** cut out and other enzymes replace damaged stretches of DNA.
- The ends of eukaryotic chromosomal DNA get shorter with each round of replication. The presence of **telomeres**, repetitive sequences at the ends of linear DNA molecules, postpones the erosion of genes. Telomerase catalyzes the lengthening of telomeres in germ cells.

? Compare DNA replication on the leading and lagging strands, including similarities and differences.

CONCEPT 16.3

A chromosome consists of a DNA molecule packed together with proteins (pp. 330–332)

- The chromosome of most bacterial species is a circular DNA molecule with some associated proteins, making up the nucleoid.
- The **chromatin** making up a eukaryotic chromosome is composed of DNA, **histones**, and other proteins. The histones bind to each other and to the DNA to form **nucleosomes**, the most basic units of DNA packing, which exist as part of the 10-nm fiber. Histone tails extend outward from each bead-like nucleosome core. Additional coiling and folding lead ultimately to the highly condensed chromatin of the metaphase chromosome.
- Chromosomes occupy restricted areas in the interphase nucleus. In interphase cells, most 10-nm fiber chromatin is loosely arranged (**euchromatin**), but some is more densely arranged (**heterochromatin**). Euchromatin, but not heterochromatin, is generally accessible for gene transcription.

? Describe the levels of chromatin packing you'd expect to see in an interphase nucleus.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- In his work with pneumonia-causing bacteria and mice, Griffith found that
 - the protein coat from pathogenic cells was able to transform nonpathogenic cells.
 - heat-killed pathogenic cells caused pneumonia.
 - some substance from pathogenic cells was transferred to nonpathogenic cells, making them pathogenic.
 - the polysaccharide coat of bacteria caused pneumonia.
- What is the basis for the difference in how the leading and lagging strands of DNA molecules are synthesized?
 - The origins of replication occur only at the 5' end.
 - Helicases and single-strand binding proteins work at the 5' end.
 - DNA polymerase can join new nucleotides only to the 3' end of a pre-existing strand, and the strands are antiparallel.
 - DNA ligase works only in the 3' → 5' direction.
- In analyzing the number of different bases in a DNA sample, which result would be consistent with the base-pairing rules?

(A) A = G	(C) A + T = G + C
(B) A + G = C + T	(D) A = C
- The elongation of the leading strand during DNA synthesis
 - progresses away from the replication fork.
 - occurs in the 3' → 5' direction.
 - produces Okazaki fragments.
 - depends on the action of DNA polymerase.
- In a nucleosome, the DNA is wrapped around
 - histones.
 - ribosomes.
 - polymerase molecules.
 - a thymine dimer.